

NOVABANK • EXECUTIVE MEMORANDUM

TO: Executive Committee & Retail Banking Leadership, NovaBank **DATE:** September 12, 2026 **FROM:** Lead Commercial Analytics Specialist
SUBJECT: Predictive Decision Framework: Deposit Retention & Margin Optimization **CODE REPO:** github.com/popicu/novabank-retention-framework

1. EXECUTIVE SUMMARY & DECISION

NovaBank faces margin pressure from untargeted outbound campaigns. The supplied dataset contains **41,188 contacts**, of which **11.27%** historically subscribed to a term deposit. Because the dataset does not contain a customer-churn outcome, term-deposit subscription is used as a proxy for proactive retention engagement. We recommend a **ranking-based Top-20% policy**: score customers using pre-call information and contact only the highest-scoring 20%.

Decision: Concentrate limited calling capacity where predicted response is highest, while monitoring conversion capture, outreach cost, campaign profit, and subgroup selection rates.

2. EMPIRICAL EVIDENCE & MODEL TRADE-OFFS

Call **duration** was excluded because it is only known after the call and would create look-ahead leakage. Models were evaluated with a stratified 80/20 train/test split. Gradient Boosting outperformed Logistic Regression on ROC-AUC, Brier score, Precision@20%, Recall@20%, and Lift@20%.

Campaign strategy	Contacts	Conversions	Precision	Cost	Gross value	Net profit
Legacy mass outreach	41,188	4,640 (100.0%)	11.27%	€617,820	€1,160,000	€542,180
Baseline — Logistic Top 20%	8,238	3,010 (65.1%)	36.67%	€123,570	€752,500	€631,430
Recommended — Gradient Boosting Top 20%	8,238	3,110 (67.0%)	37.75%	€123,570	€777,500	€653,930

Trade-off: The Top-20% policy contacts 8,238 customers, captures about 3,110 historical conversions, leaves about 1,530 historical converters uncontacted, and produces about 5,128 non-converting contacts. The economic case comes from concentrating conversion density while cutting call volume by 80%.

3. EXPLAINABILITY, GOVERNANCE & ETHICAL FAIRNESS

Permutation importance provides an auditable, model-agnostic view of drivers. Strong individual predictors include employment/economic indicators, contact channel, and campaign timing/history. The subgroup audit shows material differences in selection rates (for example, the 60+ group is selected at a much higher rate), so **fairness is not assumed or proven**. Age, marital status, and education selection rates should be monitored before and during deployment, with human review of operational policy.

4. RISK SENSITIVITIES & ASSUMPTIONS

Campaign value: €15 per call and €250 per subscription are illustrative assumptions, not values contained in the dataset. At €200 campaign value, the Top-20% policy remains more profitable than mass outreach (approximately **€498,430 vs. €310,180**).

Interest-rate stress: An illustrative 100 bps reduction is modeled as an 8.5% conversion reduction; under that scenario the Top-20% policy remains profitable at approximately **€587,930**. This is a sensitivity assumption, not a causal estimate.

Contact saturation: Historical conversion rates generally decline as campaign-contact count rises (13.04% at one contact versus 4.25% at eight). This is descriptive association, not proof of causal fatigue.

5. 60-DAY PILOT & OPERATIONAL ROLLOUT PLAN

Weeks 1–2	Integrate scoring into CRM; route the highest-scoring 20% to the outbound team.
Weeks 3–6	Run a randomized 4,000-customer pilot: 2,000 model-targeted vs. 2,000 control.
Weeks 7–8	Scale only if incremental conversion lift and campaign economics are validated; continue subgroup and drift monitoring.

Primary causal KPI: incremental conversion lift versus randomized control. **Supporting KPIs:** Precision@20% ≥ 35%, Recall@20%, and Return on Campaign Spend ≥ 5×. The model is a ranking aid, not an automatic customer-decision system.

AI assistance supported coding scaffolding, chart formatting, and Markdown structure; analytical framing, decision policy, assumptions, and leakage control were human-directed. Reproducible artifacts include main.py, the notebook, requirements.txt, and diagnostic/output files.